Ordering food at the self ordering kiosk

Project documentation

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Note!

The project in question will be reffered by multiple names such as: Self-Ordering Kiosk for ULBS Canteen, Self ordering kiosk system, kiosk, etc.

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1. Project description

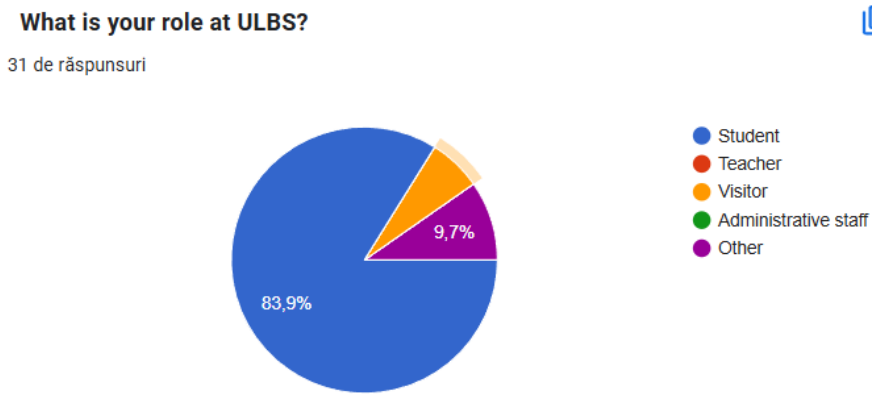
This project involves developing a self-ordering kiosk software for the ULBS food court. The kiosk will allow students and professors to place food orders from a user-friendly interface, apply eligible discounts, process payments, and notify users when their order is ready for pickup. The software will also include an admin interface for managing menu items, seamless operations and customer satisfaction.

Motivation:

The modern dining experience is increasingly shaped by technology, with self-ordering kiosks emerging as an efficient solution for improving customer satisfaction. The Self-Ordering Kiosk for ULBS Canteen project is driven by data highlighting issues such as long wait times, reliance on manual order-taking, and order inaccuracies. Customer feedback from similar implementations emphasizes the need for automated systems to streamline the ordering process.

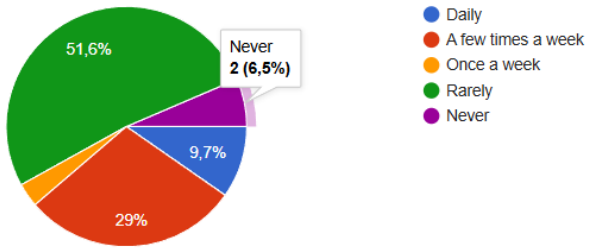
Analysis of foot traffic and order volume at ULBS Canteen reveals congestion that can be alleviated by self-service kiosks, which reduce staff workload and enhance accuracy. This project aims to develop a user-friendly kiosk solution to improve efficiency and customer experience, transforming the dining process for ULBS students, faculty, and staff.

Here are the answers of relevant questions to a survey completed by a sample of individuals who are or are not part of „Lucian Blaga" University of Sibiu:



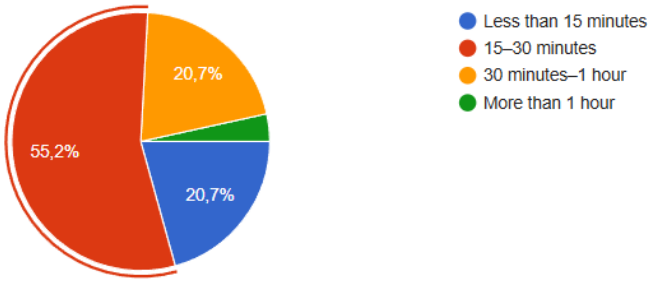
How often do you visit the ULBS food court?

31 de răspunsuri



How much time do you typically spend in the food court during each visit?

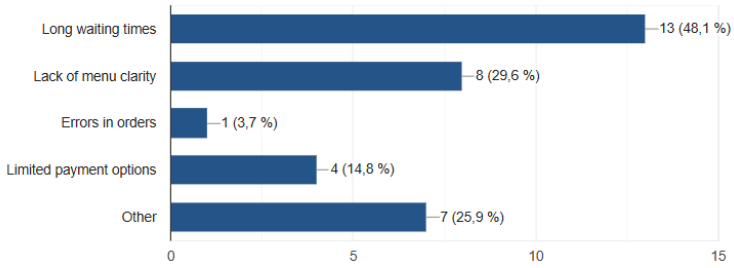
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What are the main challenges you face in the current ordering process? (select all that apply)

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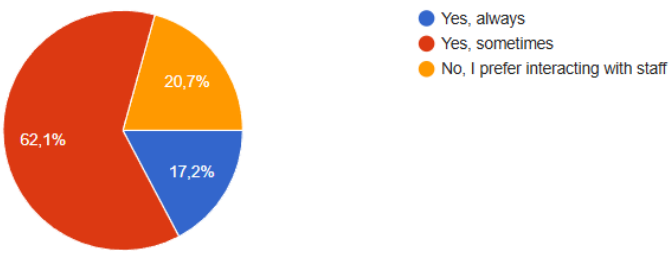
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Would you prefer to use a kiosk rather than interacting with staff to place your order?

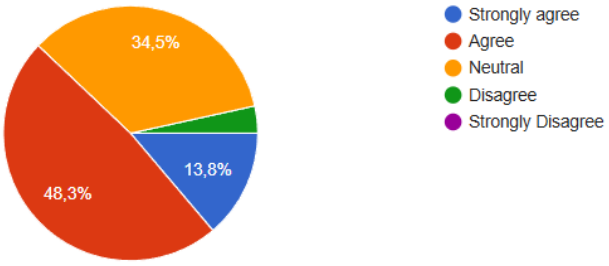
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Do you think a self-ordering kiosk will make the ordering process more convenient?

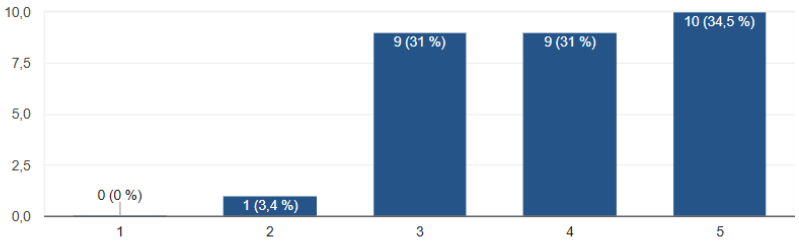
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How interested are you in using a self-ordering kiosk at the ULBS food court?

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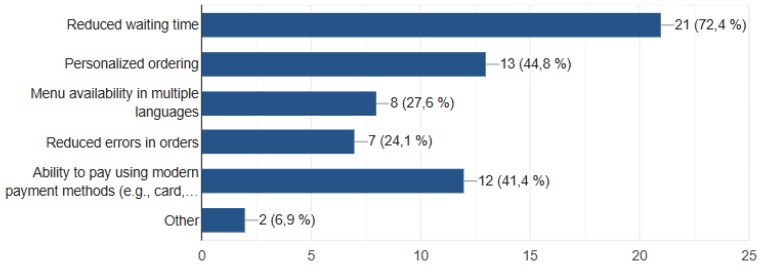
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What do you think are the main advantages of a self-ordering kiosk? (select all that apply)

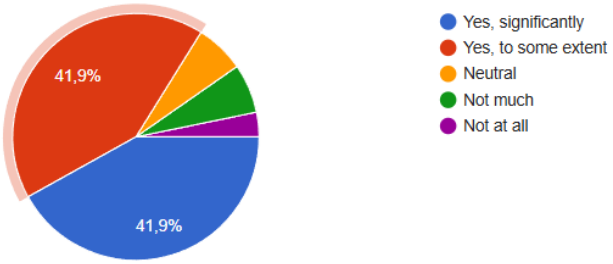
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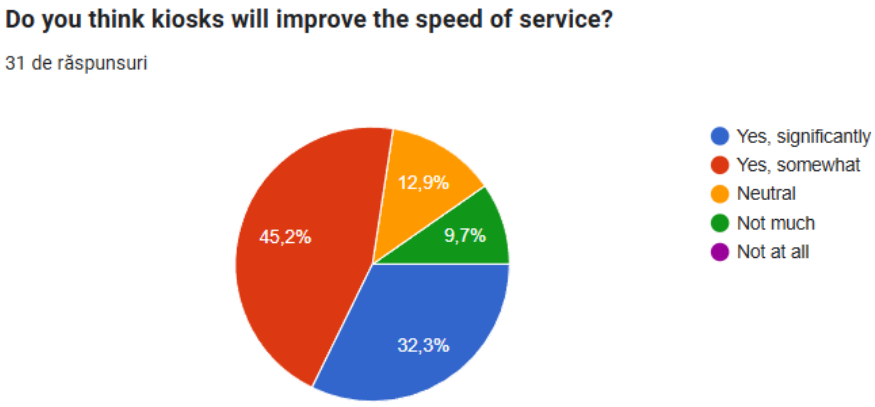
29 de răspunsuri



Do you think implementing a self-ordering kiosk would improve the overall experience at the ULBS food court?

31 de răspunsuri





In conclusion, the statistical data clearly indicates that a significant portion of respondents experience issues with long waiting times, underscoring the need for improved efficiency. Furthermore, the majority express strong support for the implementation of a self-ordering kiosk, highlighting its potential to streamline the ordering process and enhance the overall customer experience.

2. Business objectives

2.1 Enhance Customer Experience:

a) Reduce the average time spent placing an order by 50%.
b) Ensure at least 90% user satisfaction in surveys conducted after the kiosk implementation.

2.2 Increase Operational Efficiency:

a) Decrease order errors by 95% through automation of the ordering process.
b) Reduce food court staff involvement in order-taking by 30%, allowing them to focus on other tasks such as food preparation and service.

2.3 Boost revenue

a) Achieve a 20% increase in total sales by improving customer flow and reducing waiting times.

2.4 Cost Savings

a) Save on labor costs by reducing reliance on staff for order-taking during peak hours.

3. Success criteria

3.1 Customer adoption

a) At least 80% of food court customers use the kiosk for ordering within the first six months.

3.2 Revenue and Efficiency Gains

a) Demonstrate a 20% increase in daily food court sales compared to pre-kiosk implementation data.
b) Reduce average waiting time during peak hours from 10 minutes to 5 minutes.

3.3 User Feedback:

a) Gather feedback from 500+ users, ensuring at least 90% find the system "easy to use" or "very easy to use."

4. Stakeholders

Stakeholder	Major benefits	Attitudes	Win conditions	Constrains
ULBS Canteen Manager	Improved stock management, order accuracy, and overall operational efficiency.	Supportive, but expects timely delivery and robust functionality.	Successful deployment of a reliable and easy-to-use system that integrates seamlessly with operations.	Budget and resource constraints for implementation.
Customers (Students)	Faster service, reduced waiting times, and access to discounts.	Enthusiastic if the system is user-friendly and discounts are applied correctly.	Intuitive UI, accurate discount application, and reduced average waiting times.	Adoption depends on the system's ease of use and real-time stock availability..
Customers (Proffessors)	Streamlined ordering process, less time spent in queues, and personalized discounts.	Positive, provided the system is easy to use.	Accurate application of discounts and reduced waiting time.	Potential resistance if the kiosk misidentifies or fails to apply discounts correctly.
Development Team	Opportunity to deliver a scalable and impactful solution that	Fully invested in delivering a high-quality system within the agreed	Completion of the system with no major post-deployment	Limited time and resources for development; reliance on external systems

	enhances operational efficiency.	timeline.	bugs and a positive user adoption rate.	for integration.
Food Court Staff	Reduced workload related to order-taking and more efficient operations.	Supportive if the system reduces their workload and is reliable.	Fewer customer complaints due to kiosk errors and a streamlined workflow.	Training required to familiarize staff with managing kiosk-related issues.

5. Vision

For: students, proffessors, staff, and visitors at the ULBS canteen
Who: seek a faster, more convenient, and seamless dining experience
The: Self-Ordering Kiosk System
Is an: interactive, user-friendly solution for ordering meals and beverages
That: minimizes wait times and enhances order accuracy
Unlike: traditional manual ordering systems that are prone to delays, errors, and inefficiencies
Our product: leverages intuitive technology to provide a personalized, efficient, and accessible ordering experience, aligning with modern expectations and reducing operational overhead.

6. Project scope

The main scope of the project is to create a self-ordering kiosk system for the ULBS canteen that enhances time efficiency for customers by reducing wait times and streamlines order management, allowing staff to focus on food preparation and service rather than handling orders manually.

6.1 User features (implemented):

a) Dining Options: Users can select between "Dine In" or "Takeaway".
b) Product Navigation: Users can browse items grouped into categories for easier discovery.
c) Product Details: Users can see optional images, detailed descriptions, and pricing information that help customers make informed decisions.
d) Quantity Management: Users can select the desired quantity of items and add them to their cart.
e) Order Placement: Users can place orders, which are saved in the database, and receive a notification confirming that their order (e.g.: "Order no. X ") was placed successfully.

6.2 Admin features (implemented):

a) Secure Admin Login: Admins can log into a dedicated admin page.

b) Product Management: Admins can add new items to the database and delete existing items as needed to keep the catalog updated.

6.3 Features not yet implemented:

a) Receipt Generation: Generate and provide a receipt for every successful purchase, with an option to print it.
b) Advanced Customization**:** The current implementation does not support specific customization of orders (e.g., special instructions like “extra spicy” or “no onions”).
c) Order Management: Admins should be able to view a paginated list of orders sorted by date and time.
d) Order Tracking: Customers cannot track order preparation or status updates in real time.

6.4 Planned Growth Strategies:

a) Real-time Order Tracking: Allow users to monitor the preparation and readiness of their orders.
b) Enhanced Reporting: Improve admin insights with detailed charts and filters.
c) Advanced Admin Tools: Include inventory management and low-stock alerts for better operational efficiency.

7. Assumptions and dependencies

7.1 Assumptions:

AS-1: The canteen's existing infrastructure (hardware, network, etc.) will be capable of supporting the self-ordering kiosk system without requiring significan upgrades.

AS-2: Users (students, proffessors, staff, and visitors) will be familiar with basic touchscreen interactions and will be comfortable using the self-ordering kiosks.

AS-3: The product data (including categories, descriptions, and images) will be provided by the canteen management and kept up to date.

AS-4: There will be sufficient staff available for food preparation and delivery, as the

kiosk system aims to streamline the ordering process and shift focus to cooking rather than order-taking.

7.2 Dependencies

DE-1: The kiosk system depends on a stable database to store order and product information, which must be maintained and backed up regularly.

DE-2: The project relies on good internet connection and hardware components (such as touchscreen display)

DE-3: The project relies on effective collaboration with the canteen's management and IT department to ensure smooth deployment, updates, and system maintenance.

7.3 Related projects

RP-1: Any future integration with payment systems (e.g., card or mobile payment) will affect the project's timeline and scope.

RP-2: A future inventory management system that integrates with the kiosk could provide enhanced product and order tracking capabilities, influencing the evolution of this system.

RP-3: Development of a customer account management system or mobile app for order tracking could complement and extend the functionality of the kiosk.

8. Constraints

Dimension	Constraint (State limits)	Driver (State objective)	Degree of Freedom (State allowable range)
Features	70-80% of high-priority features must be included in release 1.0	Ensure release 1.0 meets the core user needs	Can adjust non-critical features for future releases if necessary
Quality	90-95% of user	Deliver a stable	Can adjust the

	acceptance tests must pass for release 1.0; 95-98% for release 1.1	product with minimal defects	acceptance criteria or perform more tests in release 1.1
Cost	Budget overrun up to 15% acceptable without executive review	Keep project within reasonable financial limits	May require trimming features or adjusting the schedule to stay within budget
Schedule	Release 1.0 to be available by 10/1, release 1.1 by 12/1	Meet release deadlines with key features ready	Can adjust feature prioritization or consider extending deadlines for non-critical parts
Staff	Maximum team size is 6 developers + 4 testers	Work efficiently within the available team size	May require prioritizing tasks, cross-training, or outsourcing for non-core tasks

9. Milestones

Event or Deliverable	Target Date	Responsibility
Project charter approved		Project Manager, Stakeholders
Project plan completed		Project Manager, Team Leads
Project plan approved		Project Sponsor, Stakeholders
Project team assembled		Project Manager
Project execution initiated		Project Manager, Team Leads
Project execution completed		Project Manager, Team Leads
Customer acceptance		Project Manager, Customer
Project closed out		Project Manager,

		Stakeholders

10.Business risks

Risk	Probability	Impact	Mitigation
Marketplace competition	High	High	Conduct market research to identify key competitors, refine the product to ensure competitive advantage, and consider unique selling points (USPs).
Timing issues	High	High	Develop a detailed project schedule with contingency plans; continuously track progress and adjust timelines as needed.
User acceptance	Medium	High	Conduct user testing early and throughout development; include user feedback and prioritize essential features for the first release.
Implementation issues	High	High	Ensure thorough testing at every phase, create clear documentation, and provide adequate

			training for the team to handle technical challenges.
Resource limitations (staffing)	High	High	Ensure that the team has the necessary skills; consider cross-training, outsourcing, or scaling up staffing if needed during critical phases.
Business priorities change	Low	High	Maintain regular communication with key stakeholders to ensure alignment with business goals, and be flexible to adjust the project scope if necessary.
Customer dissatisfaction	Medium	High	Implement a customer feedback loop and offer rapid response to issues post-launch; monitor product performance and address complaints promptly.

11. Resources

Resource	Description and source
Human Resources	- Key Individuals: Project manager with experience in project execution and leadership. - Teams: Development, QA, project support teams.

	- Critical Skill Sets: Software development (e.g., Java, SQL) project management tools, quality assurance, business analysis, and database management. - Vendors/Subcontractors: Third-party vendor for cloud services, hardware, or specialized software tools. - Support Functions: IT for infrastructure, finance for budget management.
Funding	- Total Budget: Estimated project budget allocated for development, training, and deployment. - Source: Internal corporate funding and potential external funding based on milestones.
Physical Resources	- Office Space: Rooms dedicated for team collaboration and individual work.
Equipment and Tools	- Computers: Laptops and desktops for all team members with appropriate specifications. - Software Tools: Development environments (e.g.: IntellijIdea, MySQL Workbench), project management tools, communication tools, version control systems.
Other Resources	- Training: Scheduled sessions for team education on tools, software, and methodologies.

12. Approvals

Please indicate the decision for this project charter:

 Approved, development of detailed project plan is authorized

 Approved, project execution is authorized

 Approved, but project is on hold until future notice

 Revise charter and resubmit for approval

 Charter and project proposal are rejected

Role or title	Name and signature	Date
Project Manager		
ULBS Canteen Manager		
Project Sponsor		

13. User stories

- As a customer I can access the menu so that I can pick my order.
- As a customer I can navigate through the menu's sections so that I can decide what I want to eat.
- As a customer I can select if I want to eat in the canteen or to take away so that I can eat at home if I want to.
- As a customer, I can select the quantity of an item so that I can personalize my order based on my needs.
- As a customer, I am notified that my order was placed successfully so that I can be assured my request is being processed.
- As a customer I can see the product's details so that i can choose what I'm going to eat.
- As a customer I can introduce a discount code so that I can receive a price reduction on my order.
- A an administrator I can log safely into an administration page so that i can manage products.
- As an administrator I can delete products from a database so that I can keep the menu updated.
- As an administrator I can add products into a database so that I can ensure new items are available for customers to order.

14.Use cases

14.1 Implemented usecases:

a)

Name:	Place an order
Primary actor:	Customer
Scope:	Self ordering kiosk system
Brief:	The user place an order and recieves confirmation.
Preconditions:	The user is at the self-ordering kiosk and the system is fully operational.

	The user has basic knowledge of how to use the kiosk. The user has the ability to view and select items from the menu.
Postconditions:	The order is successfully placed and recorded in the system. The user receives a confirmation message with the order number (e.g., “Order no. X”). The user’s cart is cleared and the user is redirected to homepage.
Basic flow:	The user can choose whether they want to dine in or take away. They can navigate through the menu, select food items, and specify the quantity of each item. The user can place the order from the cart section and once the order is placed, they will be informed that order no. X has been placed successfully.

b)

Name:	Delete product(s)
Primary actor:	Administrator
Scope:	Self ordering kiosk system
Brief:	The administrator can delete one or many products at a time from the meniu.
Preconditions:	The administrator is logged into the secure admin interface. The system is operational, and the database is accessible.
Postconditions:	The selected product is deleted from the database. The product is no longer available for customers to order on the kiosk.
Basic flow:	The administrator logs into the secure admin interface, and the system validates their credentials. Once logged in, the administrator navigates to the „delete products” section, where the system displays a list of all available products. The administrator then selects the product(s) they wish to delete and click the „DELETE” button. The system removes the selected product(s) from the database and updates the menu to ensure the deleted items are no longer available for customers to order.

c)

Name:	Navigate through menu
Primary actor:	Customer
Scope:	Self ordering kiosk system
Brief:	The user navigates through the meniu in order to decide what they want to add to their order.
Preconditions:	The user is at the self-ordering kiosk and the system is fully operational. The user has basic knowledge of how to use the kiosk. The user chose if he wants to dine in or take away an the meniu is now displayed.
Postconditions:	The user has browsed the menu and knows the food items he wants to add to the order. The menu displays the available items, and the user can proceed to add them to the cart.
Basic flow:	The user approaches the self-ordering kiosk and selects whether they want to dine in or take away. Once the choice is made, the system displays the menu with food categories and available items. The user then navigates through the menu by selecting different category options. Upon selecting a category, the system shows the specific food items within that category, including product names, pictures, descriptions, and prices.

14.2 Underimplemented usecases:

a)

Name:	Apply discount based on a code
Primary actor:	Customer(student or proffessor)
Scope:	Self ordering kiosk system
Brief:	The customer who is a student or a proffessor introduces a discount code and the discount is applied to the total price.
Preconditions:	The user is at the self-ordering kiosk and the system is fully operational. The user has basic knowledge of how to use the kiosk. The products are in the cart. The user has access to a discount code.
Postconditions:	A discount is applied to the total price.

Basic flow:	The user approaches the self-ordering kiosk and adds products to the cart. Once the items are added to the cart, the system prompts the user to enter a discount code. The user enters the discount code into the designated input field on the kiosk screen, and the system verifies the code's validity. If the code is valid, the system applies the discount to the total price and displays the updated total price reflecting the discount. The user proceeds to finalize the order, with the discounted total price shown.

b)

Name:	Add product to the menu
Primary actor:	Administrator
Scope:	Self ordering kiosk system
Brief:	The administrator adds new product to the meniu.
Preconditions:	The administrator is logged into the secure admin interface. The system is operational, and the database is accessible.
Postconditions:	The product is added to the meniu. The product is available for customers to order on the kiosk.
Basic flow:	The administrator logs into the secure admin interface of the system and navigates to the „add products” section. The system prompts them to enter the product details, such as the name, description, price, image URL and category. The administrator enters the necessary information and submits the new product for addition. The system updates the database with the new product. Once added, the product becomes available on the menu and can be selected by customers on the kiosk.

14.3 Usecases that need to be implemented

a)

Name:	Update order status
Primary actor:	Kitchen staff

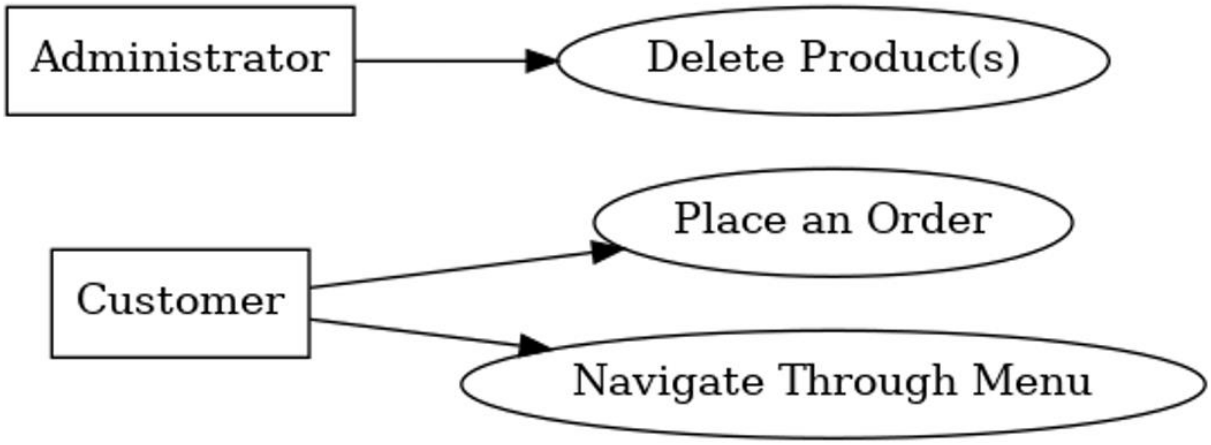
Scope:	Self ordering kiosk system
Brief:	The kitchen staff updates the status of an order to keep the customer informed and manage the food preparation process.
Preconditions:	The kitchen staff is logged into the secure admin interface. An order has been placed through the self-ordering kiosk and is in the system. The system is operational.
Postconditions:	The status of the order is updated (e.g., from "preparing" to "ready"). The updated status is reflected on the customer's screen.
Basic flow:	The kitchen staff logs into the secure admin interface using their kitchen staff credentials. Once logged in, the system displays a page with a list of current orders, including order details and the ordered products for each. When an order is complete, the kitchen staff clicks the "Ready" button next to the specific order. Once the order is marked as ready, the system updates the status of the order, and the result is displayed on the customer's kiosk screen, notifying them that their order is ready for pickup.

b)

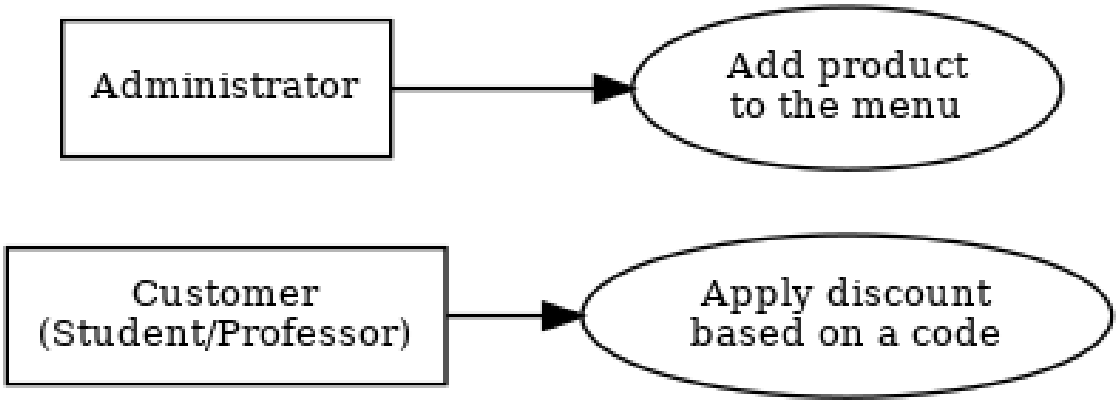
Name:	View order history
Primary actor:	Administrator, kitchen staff
Scope:	Self ordering kiosk system
Brief:	The administrator or kitchen staff views the order history to monitor and manage customer orders.
Preconditions:	The administrator or kitchen staff is logged into the secure admin interface. The system is operational, and the database is accessible. Orders have been placed through the self-ordering kiosk.
Postconditions:	The administrator or kitchen staff can view a complete list of customer orders, including order details (e.g., item names, quantities).
Basic flow:	The administrator or kitchen staff logs into the secure admin interface and navigates to the "Order History" section. The system displays a list of all past orders, showing essential information such as order numbers,

	item details, and order status. The user can select a specific order to view more details, including items, quantities, prices, and customer information. The system allows the user to filter or sort orders by date, customer, or status to better manage and review the order history. The order history can be used by the administrator or kitchen staff for reporting purposes or to fulfill outstanding orders efficiently.

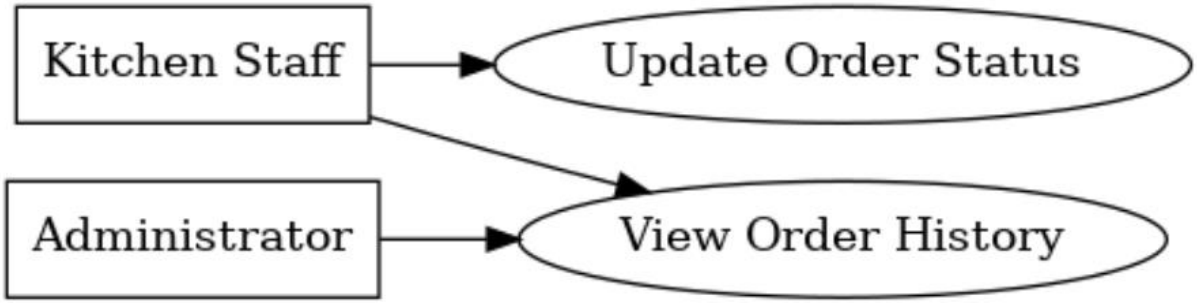
14.4 Implemented usecases diagram



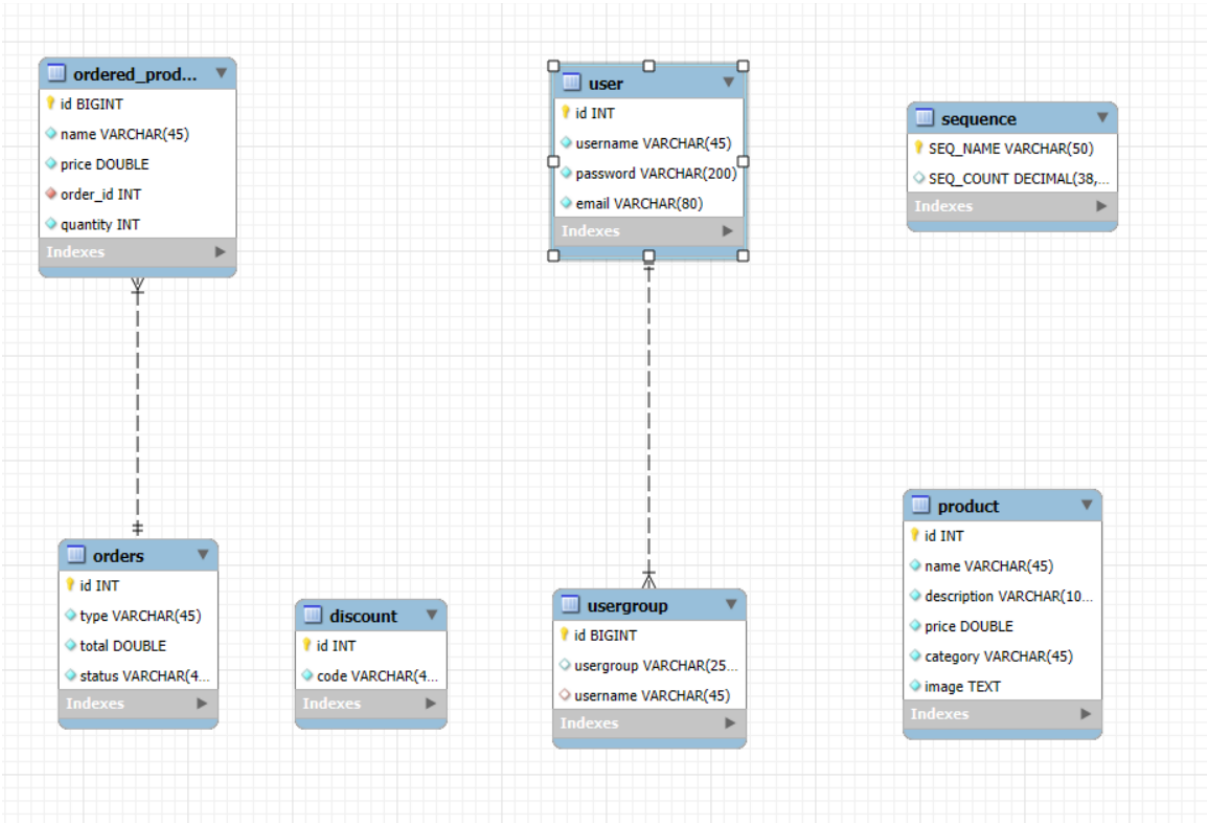
14.5 Underimplemented usecases diagram



14.6 Diagram for usecases that need to be implemented



15. Database schema



16. System sequence diagrams

